

★ MONTH 3 – COMPLETING CLASS 12 SYLLABUS

■ PHYSICS (Class 12)

1. Dual Nature of Radiation & Matter

Subtopics:

1. Photoelectric Effect
 - Experimental setup
 - Einstein's Photoelectric equation
 - Stopping potential
 2. Wave-Particle Duality
 - de Broglie wavelength
 3. Electron Emission
 4. Energy Quantization
 5. Applications of Photoelectric Effect
-

2. Atoms & Nuclei

Subtopics:

Atoms

1. Thomson, Rutherford Model
2. α -particle Scattering
3. Bohr's Model
4. Energy Levels & Spectra
5. Hydrogen Line Spectrum

Nuclei

6. Nuclear Structure & Density
 7. Radioactivity (α , β , γ decay)
 8. Radioactive Law $\rightarrow N = N_0 e^{-\lambda t}$
 9. Half-life & Mean Life
 10. Nuclear Fission & Fusion
 11. Mass Defect & Binding Energy
-

3. Semiconductor Electronics

Subtopics:

1. Intrinsic & Extrinsic Semiconductors
 2. p-type & n-type
 3. p–n Junction Diode
 - I-V Characteristics
 4. Zener Diode (Breakdown, Use as Voltage Regulator)
 5. Rectifiers
 - Half-wave
 - Full-wave
 6. Logic Gates (NOT, AND, OR, NAND, NOR)
 7. Truth Tables
 8. Basic Digital Circuits
-



CHEMISTRY (Class 12)

1. d– and f–Block Elements

Subtopics:

1. General Properties of Transition Metals
 2. Trends in Transition Metals
 3. Colour, Complex Formation, Magnetic Properties
 4. Interstitial Compounds
 5. Lanthanides → Properties & Trends
 6. Actinides → Properties
 7. Important Compounds (KMnO_4 , $\text{K}_2\text{Cr}_2\text{O}_7$)
-

2. Coordination Compounds

Subtopics:

1. Ligands & Coordination Number
2. Nomenclature (Very Important)
3. Werner's Theory
4. Chelation
5. Bonding (VBT, CFT)
6. Crystal Field Splitting in
 - Octahedral

- Tetrahedral

7. Colour & Magnetism
 8. Isomerism (Ionization, Linkage, Geometrical, Optical)
-

3. Haloalkanes, Haloarenes / Alcohols, Phenols, Ethers

(These two chapters are usually clubbed)

Subtopics:

Haloalkanes & Haloarenes

1. Preparation
2. Physical & Chemical Properties
3. Nucleophilic Substitution (SN1, SN2)
4. Elimination Reactions
5. Reactivity Order
6. Environmental Effects (CFCs)

Alcohols, Phenols, Ethers

7. Preparation
 8. Acidity of Alcohols/Phenols
 9. Reactions (Oxidation, Esterification, Dehydration)
 10. Williamson Synthesis
 11. Electrophilic Substitution in Phenol
-

MATHEMATICS (Class 12)

1. Integration

Subtopics:

1. Indefinite Integration Basics
 2. Standard Integrals
 3. Integration by Substitution
 4. Integration by Parts
 5. Partial Fractions
 6. Definite Integrals
 7. Properties of Definite Integrals
 8. Area Under Curve
-

2. Differential Equations

Subtopics:

1. Order & Degree
 2. Formation of Differential Equation
 3. General & Particular Solutions
 4. Variable Separation Method
 5. Homogeneous Differential Equations
 6. Linear Differential Equations
 - $\frac{dy}{dx} + Py = Q$
 7. Applications (Growth/Decay Models)
-

3. Vectors + 3D Geometry

Vectors Subtopics:

1. Vector Basics
2. Dot Product
3. Cross Product
4. Scalar Triple Product
5. Vector Projections

3D Geometry Subtopics:

6. Direction Cosines & Ratios
 7. Equation of Line
 - Vector form
 - Cartesian form
 8. Angle Between Two Lines
 9. Plane equation
 10. Angle Between Line & Plane
 11. Distance (Point → Line, Point → Plane)
-



1-MONTH (4-WEEK) PLAN: Month 3

Week 1

Physics: Dual Nature

Chemistry: d & f Block

Math: Integration

Week 2

Physics: Atoms

Chemistry: Coordination Compounds

Math: Integration (Advanced) + Start Differential Equations

Week 3

Physics: Nuclei + Semiconductors

Chemistry: Haloalkanes / Alcohols

Math: Differential Equations (Finish)

Week 4

Math: Vectors + 3D Geometry

Chemistry: Full revision

Physics: PYQs + Important numericals

Mixed mock tests (Boards + JEE)

After Month 3

- ✓ Entire Class 12 Syllabus Completed
- ✓ 90% Board Syllabus Done
- ✓ Half of JEE Main Chemistry Strong
- ✓ Physics Numericals solid
- ✓ Maths Calculus (heavy weightage) completed